

Phonetics and Phonology ENG507(Finals) 2nd Batch

Q. What are gestures?

In phonetics and phonology, speech sounds (segments) using basic units of contrast are defined as gestures.

Q. What are Articulatory Targets?

All consonants have a fixed articulatory target which is realised at a single precise place of articulation. The articulatory target for a particular consonant is known as its target for a particular consonant is known as its articulatory locus.

The traditional terms which are used for all the places of articulation are not just names for particular locations (on the roof of the mouth). They should be thought of as names for articulatory targets.

Q. Write down types of Articulatory Gestures.

i. Bilabial gesture – (e.g., stops and nasal: p, b, m). The symbols for the voiceless and voiced bilabial fricatives are [ɸβ]. These sounds are pronounced by bringing the two lips nearly together, so that there is only a slit between them.

ii. Labiodental fricatives – [f, v]. In English, a labiodental nasal, [ɱ], may occur when /m/ occurs before /f/, as in *emphasis* or *symphony*.

iii. Dental - e.g. dental fricatives [θ, ð] but there are no dental stops, nasals, or laterals except allophonically realized (before [θ, ð] as in *eighth*, *tenth*, *wealth*). Many speakers of French, Italian, and other languages (such as Urdu, Pashto and Sindhi) typically have dental stops such as [t̪ d̪].

iv. Alveolar are very common targets and stops, nasals, and fricatives all occur in English and in many other languages at alveolar as a target of articulatory gestures (e.g., t, d, n, l, r., etc.).

v. Retroflex is very common sound in many Pakistani languages which is made by curling the tip of the tongue up and back so that the tongue tip moves during the retroflex sounds such as [ɳ, ɳ̌, ɳ̍, ɳ̎, ɳ̏].

vii. Palato-alveolar and **palatal** are also possible articulatory gestures commonly found in world languages. Similarly, velar sounds found in Urdu and other Pakistani languages need to be mentioned here including [x, ɣ] which are velar fricatives. The gestures for pharyngeal (such as Arabic pharyngeal fricative [ʕ]) and epiglottal sounds (such as epiglottal fricative [ʕ̥]) involve pulling the root of the tongue or the epiglottis back toward the back wall of the pharynx.

Q. What are types of stops?

The following table shows stop sounds found in various languages:

Description	Symbol	Language
1. Voiced	b	English and other languages
2. Voiceless unaspirated	p	-do-
3. Aspirated	p ^h	Sindhi and many other Pakistani languages
4. Murmured (breathy)	b ^ɦ	Sindhi
5. Implosive	ɓ	Sindhi
6. Laryngealized (creaky)	ɓ̥	Hausa
7. Ejective	kʼ	Hausa
8. Nasal release	dn	Russian

Phonetics and Phonology ENG507(Finals) 2nd Batch

9. Prenasalized	nd	Swahili
10. Lateral release	tl	Navajo
11. Ejective lateral release	tl'	Navajo
12. Affricate	ts	German
13. Ejective affricate	ts'	Navajo

Describe nasals.

Like stops, nasal can also occur voiced or voiceless (for example, in Burmese, Ukrainian and French) though in English and other most languages nasals are voiced. As voiceless nasals are comparatively rare, they are symbolized simply by adding the voiceless diacritic [_h] under the symbol for the voiced sound. There are no special symbols for voiceless nasals and it is written as /_hm/ - a combination of the letter for the voiced bilabial nasal and a diacritic indicating voicelessness.

Q. Explain Fricatives

Fricative as an articulatory gesture may be divided into voiced or voiceless sounds but we can also subdivide fricatives in accordance with other aspects of the gestures that produce them. For example, some authorities have divided fricatives into sounds such as [s], in which the tongue is grooved so that the airstream comes out through a narrow channel, and those such as [θ], in which the tongue is flat and forms a wide slit through which the air flows. On the other hand, a slightly better way of dividing fricatives is to separate them into groups on a purely auditory basis.

Q. Define sibilant sounds.

The fricatives [s, z, ʒ, ʃ] are called sibilant sounds. They have more acoustic energy—that is, greater loudness—at a higher pitch than the other fricative sounds.

Q. Where are Trills, Taps and Flaps found?

In approximants, trills, taps and flaps are commonly found with different articulatory gestures in the world languages. These languages vary not only in terms of the nature of the sounds but they also vary in terms of the length of the sound.

Q. Write down various types of approximant sounds.

The following table covers various types of these approximant sounds found in the world languages:

Voiced alveolar trill	r	Spanish
Voiced alveolar tap	ɾ	Spanish
Voiced retroflex flap	ɽ	Hausa
Voiced alveolar approximant	ɹ	English
Voiced alveolar fricative trill	ʀ	Czech
Voiced uvular trill	R	French
Voiced uvular fricative or approximant	ʁ	Parisian French
Voiced bilabial trill	β	Kele
Voiced labiodental flap	*	Margi

Q. What are Laterals? What are forms of laterals?

Laterals are important articulatory gestures. Laterals are usually presumed to be voiced approximants unless a specific statement to the contrary is made.

- a lateral stop

Phonetics and Phonology ENG507(Finals) 2nd Batch

- a lateral fricative
- lateral approximant

Q. Write a note on different types of sounds.

There are 13 types of stop sounds (i.e., b, p, p^h, b^h, ɓ, k', dn, nd, tɬ, tɬ', ts, ts') and nine types of trill, tap and flap (together one category of approximants) sounds (i.e., r, ɾ, ʀ, ɽ, ɻ, ʁ, B, *) with a range of similarities in terms of their articulatory gestures.

Q. Write a detailed note on Acoustic Phonetics.

Acoustics is the study of the physics of the speech signal (i.e., when speech sound travels through the air in a wave form) from the speaker's mouth to the hearer's ear through vibrations. In acoustics, it is possible to measure and analyze these vibrations (physical properties) by mathematical techniques usually by using specially-developed computer software to produce spectrograms of speech.

Acoustic phonetics also studies the relationship between activity in the speaker's vocal tract and the resulting sounds by involving physics, computer, statistics and a number of experiments in lab. Thus, the analysis of speech using the expertise available in acoustic phonetics is claimed to be more objective and scientific than the traditional auditory method which mostly depends on the reliability of the trained human ear.

It also involves advanced level speech software for analyzing sound differences (in terms of pitch, loudness and quality) and distinguishes largely among speech sounds by giving the detailed composition of energy (e.g., frequency on spectrum).

Q. Describe in detail Acoustic Analysis.

According to the experts of speech sounds (phoneticians), acoustic analysis can provide a clear, objective datum for the investigation of speech – the physical 'facts' of utterance. In other words, acoustic evidence is often referred to when one wants to support an analysis being made in articulatory or auditory phonetics. Acoustic analysis not only gives us the features of a sound but also tells us about the duration or length of a speech sound. For such an analysis, we need to carefully know about material and procedure of recording. Thus, acoustic analysis describes the durational characteristics, articulatory properties and phonetic differences through physiological measurement.

Q. What is Acoustic Analysis of Vowels?

The experts of phonetics are particularly interested in analyzing vowels acoustically. They describe vowels in terms of numbers (in a language that how many vowels are possible). It is also possible to analyze vowel sounds so that the measurement of the actual frequencies of the formants (the formant structure of the vowels of a language) is taken.

Q. Source Filter Theory of Speech Production

It is a model of speech (e.g., vowel) production. According to this theory, source refers to the waveform of the vibrating larynx. Its spectrum is rich in harmonics, which gradually decrease in amplitude as their frequency increases. The various resonance chambers of the vocal tract, especially the movements of the tongue and lips, act on the laryngeal source in the manner of a filter, reinforcing certain harmonics relative to others. Thus the combination of these two elements (larynx as source and cavity as filter) is known as the source-filter model of speech (e.g., vowel) production.

Q. Define formants.

The overtones are called formants, and the lowest three formants distinguish vowels from each other.

Phonetics and Phonology ENG507(Finals) 2nd Batch

Q. Explain in detail the Source – Filter Mechanism.

In this theory, the tract is represented using a source-filter model and several devices have been devised to synthesize speech in this way. The idea is that the air in the vocal tract acts like the air in an organ pipe, or in a bottle. Sound travels from a noise-making source (i.e., the vocal fold vibration) to the lips. Then, at the lips, most of the sound energy radiates away from the lips for a listener to hear, while some of the sound energy reflects back into the vocal tract. The addition of the reflected sound energy with the source energy tends to amplify energy at some frequencies and damp energy at others, depending on the length and shape of the vocal tract. The vocal folds (at larynx) are then a source of sound energy, and the cavity (vocal tract - due to the interaction of the reflected sound waves in it) is a frequency filter altering the timbre of the vocal fold sound. Thus this same source-filter mechanism is at work in many musical instruments. In the brass instruments, for example, the noise source is the vibrating lips in the mouthpiece of the instrument, and the filter is provided by the long brass tube.

Q. Write a detailed note on the Tube Models

The formants that characterize different vowels are the result of the different shapes of the vocal tract. Any particle of air, such as that in the vocal tract or that in a bottle, will vibrate in a way that depends on its size and shape. Remember that the air in the vocal tract is set in vibration by the action of the vocal folds (in larynx). Every time the vocal folds open and close, there is a pulse of acoustic energy (activation). Irrespective of the rate of vibration at source (of the vocal folds), the air in the vocal tract will resonate at these frequencies as long as the position of the vocal organs remains the same. Because of the complex shape of the filter (tract), the air will vibrate in more than one way at once. So, the relationship between resonant frequencies and vocal tract shape is actually much more complicated than the air in the back part of the vocal tract vibrating in one way and the air in other parts vibrating in another. In most voiced sounds, three formants are produced every time the vocal folds (source) vibrate. The resonance in the vocal tract (filter) is independent of the rate of vibration of the vocal folds (source). In other words, the vocal folds may vibrate faster or slower, giving the sound a higher or lower pitch, but the formants will be the same as long as the position of the tube (vocal tract) is the same.

Q. What is Perturbation Theory?

The theory of perturbation says that with the acoustic effect of constriction at the lips, we can predict the formant frequency differences between rounded and unrounded vowels. Keeping in mind this modification in the size and nature of vocal tract (for specific vowel sounds), we can estimate how this perturbation theory works. So for each formant, there are locations in the vocal tract where constriction will cause the formant frequency to rise, and locations where constriction will cause the frequency to fall.

Q. Explain Perturbation Theory.

According to perturbation theory, resonance occurs in a uniform tube where one end is closed and the other end is open. This theory tells us whether each resonance frequency increases or decreases when a small modification occurs in the diameter at a local region of the tube (tract). As a result:

- the resonance frequency of the particular resonance mode decreases when a constriction is located at an anti-node of that resonance mode; and
- the resonance frequency of the particular resonance mode increases when a constriction is located at a node of that resonance mode.

Keeping in mind this idea of perturbation theory, we can derive that the resonance frequencies will change (decrease or increase) as per the position (modification in the size and nature) of the vocal tract (tube).

Phonetics and Phonology ENG507(Finals) 2nd Batch

Q. Explain Acoustic Analysis (Vowels)

Using computer programs, we can analyze vowel sounds by showing their components through the display (spectrogram). In spectrograms, time runs from left to right, the frequency of the components is shown on the vertical scale, and the intensity of each component is shown by the degree of darkness. It is thus a display that shows, roughly speaking, dark bands for concentrations of energy at particular frequencies—showing the source and filter characteristics of speech. The first two frequencies are important here. The first formant (F1) is inversely related to the height of a vowel whereas the second formant (F2) is related to the frontness of a vowel sound. When the first two formants are taken, the vowels of a language can be plotted on a chart and the structure is very much related to the traditional description of vowel sounds.

Q. Explain Acoustics of Consonants.

The acoustic properties (structure) of consonantal sounds are usually more complicated than that of vowels. Usually, a consonant can be said to be a particular way of beginning or ending a vowel sound because during the production of a consonant there is no distinguishing feature prominently visible. There is virtually no difference in the sounds during the actual closures of voiced stops [b, d, g], and absolutely none during the closures of voiceless stops [p, t, k], because there is only silence at these points. Each of the stop sounds conveys its quality by its effect on the adjacent vowel. There are some consonantal sounds which have vowel like structure; therefore, their acoustic features are somehow similar to vowels (in the case of nasal consonants, approximants and glides) but most of the consonants have totally different acoustic features.

Q. Write a note on the importance of learning and using spectrogram.

1. Using Praat (or any other software) and spectrogram is particularly useful when a researcher is working on a problem related to the nature (physical properties) of a sound (e.g., is it a phoneme or allophone?).
2. It increases our understanding of the speech sounds and their behavior in different forms (in isolation or as the part of connected speech).
3. Practice on spectrogram gives us the opportunity to learn about the characteristics of speech sounds.
4. It is also important for experts who are working on phonetic aspects of speech as signal processing.
5. These are also used as the part of techniques in speech recognition.
6. Spectrograms enable us to explore the complex nature of speech structure (as the part of spoken language).
7. Spectrograms are the part of techniques used in machine translation.

Q. Write a detailed note on the useful techniques for interpreting Spectrograms

These are the following helpful points to use in interpreting spectrograms:

1. Start analyzing sounds one by one by keeping in mind the individual characteristics of sounds as a class.
2. Carefully see the overall structure, especially the frequency scale.
3. While interpreting consonants, also analyze the behavior of the adjacent vowels.
4. Pay more attention to the first two formants (especially for vowels).
5. Watch for a burst and aspiration in stop sounds.
6. Remember that in vowels the first formant is inversely related to the height of a vowel (the lower is F1, the higher is the vowel) and F2 is related to the degree of backness of the vowel.

Phonetics and Phonology ENG507(Finals) 2nd Batch

7. It is, of course, also possible to tell many other things about the manner of articulation from the spectrograms of various sounds (e.g., one can usually see whether a stop has been weakened to a fricative, or even to an approximant in some cases). Similarly, the process of affrication (of a stop) can also be seen on many occasions. Trills can be separated from taps and flaps, and voiced sounds from voiceless ones.

Q. Describe individual differences.

Following points are important in this regard:

1. Spectrograms show relative quality (e.g., a particular speaker may have a higher level of vowels).
2. The formant plots (of an average speaker) may be compared with the formant plots of a particular speaker.
3. Similarly, when two different speakers record their sets of vowels with the same phonetic quality, their relative positions on a formant chart will be similar, but the absolute values of the formant frequencies will differ.
4. In such case, the absolute values of the vowels will be important. But remember that it is a complex issue and no simple (or single) technique is useful.
5. One very important strategy may be to use the values of F4 (which may be the indicator of individual head size). Thus F4 may be studied in connection with first three formants for further evaluation. Remember that F4 for other sounds will also be required for such a comparison.

Q. Write a detailed note on vowels and vowel-like articulation

The fundamental distinction between consonant and vowel sounds is that vowels make the least obstruction to the flow of air. In addition to this, vowels are almost always found at the center of a syllable, and it is very rare to find any sound, other than a vowel which can stand alone as a whole syllable. Phonetically, each vowel has a number of features (properties) that distinguish it from other vowels. These include; firstly, the shape of the lips (lip-rounding), rounded (for sounds like /u:/ vowel), neutral (as for ə - schwa sound) or spread (as in /i:/ sound in word like sea or – when photographers traditionally ask you to say “cheese” /tʃi:z/ in order to make you look smiling. Secondly, part of the tongue - the front, the middle or the back of the tongue may be raised, giving different vowel qualities: compare /æ/ vowel (as in word ‘cat’) as a front vowel, with the /ɑ:/ vowel (as in ‘cart’) which is a back vowel. Thirdly, the tongue (and the lower jaw) may be raised ‘close’ to the roof of the mouth (for close vowels. e.g. /i:/ or /u:/), or the tongue may be left ‘low’ in the mouth with the jaw comparatively ‘open’ (as for open vowels e.g., /a:/ and /æ/).

Q. Explain in detail the cardinal vowels.

In order to classify vowels (independent of the vowel system of a particular language), the English phonetician Daniel Jones introduced a system in early 20th century and worked out on a set of vowels called the “cardinal vowels” comprising of eight vowels to be used as reference points (so that other vowels could be related to them like the corners and sides of a map).

Cardinal vowel system is a chart or four-sided figure (the exact shape of which has been changed from time to time), with eight corners as can be seen on the IPA chart from IPA website. It is a diagram to be used both for rounded and unrounded vowels, and Jones proposed that there should be a primary and a secondary set of cardinal vowels. The primary set includes eight vowels in total (from 1 to 8); the front unrounded vowels [i, e, ε, a], the back unrounded vowel [ɑ] and the rounded back vowels [ɔ, o, u].

Q. Write a note on secondary cardinal vowels

Phonetics and Phonology ENG507(Finals) 2nd Batch

Further on explaining vowels (particularly, the vowels of Non-European languages), a set of secondary cardinal vowels (as a precise set of references) was introduced by the same British phonetician Daniel Jones (1881-1967). Secondary cardinal vowels are easy to understand in connection with the primary cardinal vowel system. The main difference between primary and secondary cardinal vowels is related to lip-rounding as in some languages the feature of lip-rounding is possible for front vowels. By reversing the lip position (in comparison with primary cardinal vowels), the secondary series of vowel types is produced (e.g., rounding the lips for the front vowels).

Q. Draw a comparing between primary vs secondary cardinal vowels

Secondary cardinal vowels from 9 to 13 [y, ø, œ, æ, ʊ] are the rounded counterparts of primary cardinal vowels from 1 to 5 [i, e, ɛ, a, ɑ]. Similarly, secondary cardinal vowels from 14 to 16 (ʌ, ɤ, ɯ) are the unrounded equivalents of primary cardinal vowels from 6 to 8 [ɔ, o, u] respectively. Two further cardinal vowels (17 and 18 symbolized by [ɨ] and [ɘ]) represent the highest point at the center where the tongue can possibly reach. The entire vowel system (of human languages) is usually shown in the form of the cardinal vowel diagram (resembling human tongue and divided into eight corners) or cardinal vowel quadrilateral. The aim is to give an approximate configuration of the degree and direction of tongue movement involved in the vowel production. These diagrams have been successfully used for the description of vowel system available in the dialect and languages of the world.

Q. Write a detailed note on vowels in other accents of english

Vowel sounds in various accents of English are interesting for a number of reasons. They provide solid basis for comparisons and contrasts. For example, the accent of American (newscasters) English has represented fairly conservative difference (in the first two formants) with Californian English. The Californian English does not maintain a contrast between the vowels in cot and caught (they are both spoken as the same [ɑ]). Moreover, the Californians have a higher vowel (lower first formant) in [eɪ] than in [i]. Their high back vowels seem more towards front as they have a higher second formant. Among other differences, vowel /ʊ/ is often pronounced with spread lips in this variety of English. Similarly, in a number of northern cities in the United States (e.g., Pittsburgh and Detroit), [æ] is spoken very close to [ɛ] (as raised with decreased F1). These are some of the examples of differences found in various varieties of English.

Q. Write a note on vowels in BBC English

While discussing vowels in English, we can explore vowels in BBC accent for a number of reasons. The first and the foremost comparison may be with the American English (or varieties of American English). The vowels of BBC accent are different in both number (20 in total – both pure vowels and diphthongs) and quality. British English speakers distinguish the vowel [ʌ] in cut from the vowel [ɜ] in curt. It does not have any r-coloring (rhotacization) mainly by the frequency of the first formant. Moreover, as the main feature of BBC English (to be noted here) is the distinction between the three back vowels [a] as in father and cart, [ɑ] as in bother and cot, and [ɔ] as in author and caught. BBC vowels are really interesting in making it a standard variety of English.

Q. Describe vowels in other languages

Variations among human languages are not limited only to the number of vowels but also to the quality and features. Spanish has a very simple system contrasting only five vowel sounds [i, e, a, o, u]. It is important to remember that these symbols do not have the same values as English or their descriptions of cardinal vowels. Japanese also has a set of five vowels [i, e, a, o, u] but very different in a narrower transcription than that of Spanish. Similarly, Danish has different vowels and their qualities as it contrasts three front rounded vowels which may occur both in long or short forms. Focusing on Pakistani regional languages, Urdu and Punjabi have very different nasal vowels (the

Phonetics and Phonology ENG507(Finals) 2nd Batch

nasality is phonemic here). Urdu has some seven nasal vowels such as /he/ vs. /hẽ/ or /hi:/ vs. /hĩ:/ as in word /nahĩ:/ meaning 'no'. Nasalization in vowels is a common feature of Indo Aryan languages.

Q. What is Advanced Tongue Root (ATR)

Normally, while discussing the degree of variation in vowel sounds, three types of features are given (i.e., height of tongue, backness of tongue and lip rounding) which cover the major variation in world's languages. But this description does not cover all types of variation in vowel quality. One out of such variations is advanced tongue root (ATR) which is found in Akan language spoken in Ghana. Actually, vowels produced with ATR involve the furthest-back part of the tongue, opposite to the pharyngeal wall, which is not normally involved in the production of speech sounds - also called the radix (articulations of this type may, therefore, be described as radical). ATR (a kind of articulation in which the movement of the root of tongue expands the front-back diameter of the pharynx) is used phonologically in Akan (and some other African languages) as a factor in contrast of vowel harmony. The opposite direction of movement is called retracted tongue root (RTR). ATR is thus related to the size of pharynx – making the pharyngeal cavity different: creating comparatively large (+ATR: root forward and larynx lowered) and small pharyngeal cavity (-ATR: no advanced tongue root). Akan contrasts between two sets of vowels +ATR and -ATR.

Q. What are rhotacized vowels

In the description of vowel quality, rhotacization (or rhotacized vowel) is a term which is used in English phonology referring to dialects or accents where /r/ is pronounced following a vowel, as in words 'car' and 'cart'. Thus varieties of English are divided on the basis of this feature - varieties having this feature are rhotic (in which /r/ is found in all phonological contexts) while others (not having this feature) are non-rhotic (such as Received Pronunciation where /r/ is only found before vowels as in 'red' and 'around'). Similarly, vowels which occur after retroflex consonants are sometimes called rhotacized vowels (they display rhotacization).

Q. Write a note on nasalization in vowels

The speakers of Urdu, Punjabi and many other Pakistani regional languages learn to produce a variety of nasal vowels as the part of their mother tongue and face no issue in learning nasalization in vowels. However, for the speakers of other languages (such as English which does not have nasal vowels) have to learn this feature of vowels by starting saying the low vowel /æ/ as in man and by keeping the soft palate lowered. Many languages have contrasts between nasal and oral vowels including French and Urdu. Urdu and Punjabi have many nasal vowels. Urdu has seven nasal vowels such as /he/ (meaning 'is') vs. /hẽ/ (meaning 'are'). Nasalization in vowels is a common feature of Indo Aryan languages. In IPA chart, the diacritic used for nasalization is the symbol [~] called tilde (used above the phonetic symbol to show the nasality).

Q. What are semivowels

Most of the world languages contain a class of sounds that functions in a way similar to consonants but is phonetically similar to vowels (e.g., in English, /w/ and /j/ as in 'wet' and 'yet'). When they are used in the first part of syllables (at onset), they function as consonants. But if they are pronounced slowly, they resemble (in quality) with the vowels [u] and [i] respectively. These sounds are called semivowels which are also termed as approximants today. In French there are three semivowels (i.e., in addition to j and w there is another sound symbolized /ɥ/ and is found in initial position in the word like 'huit' /ɥit/ (eight) and in consonant clusters such as /fruɥ/ in /fruɥi/ ('fruit'). The IPA chart also lists a semivowel corresponding to the back close unrounded vowel /u/. Like the others, this is classed as an approximant.

Q. Explain secondary articulatory gestures (SAG)

Phonetics and Phonology ENG507(Finals) 2nd Batch

‘Secondary’ articulation is an articulatory gesture with a lesser degree of closure occurring at approximately the same time as another (primary) gesture. It is different than co-articulation which is at the same time and of the same value (taking place as an equal level gesture).

Q. What are types of secondary articulations?

They can usually be described as added vowel-like articulations including; ‘palatalization’ (adding a high front tongue gesture as in sound /i/), velarization (raising of the back of the tongue), pharyngealization (it is the superimposition of the narrowing of the larynx) and labialization (the addition of lip-rounding).

Q. Explain suprasegmental (SS) features

Supra means ‘above’ or ‘beyond’ and segments are sounds (phonemes). Suprasegmental is a term used in phonetics and phonology to refer to a vocal effect (such as tone, intonation, stress, etc.) which extends over more than one sound (segment) in an utterance.

Major suprasegmental features include pitch, stress, tone, intonation or juncture. Phonological studies can be divided into two fields: segmental phonology and suprasegmental phonology.

Q. Explain Syllable

In a simple way of defining the term, syllables are the parts of word (in which a word is further divided into parts), for example, *mi-ni-mi-za-tion* or *sup-re-seg-men-tal*. Phonetically, we can observe that the flow of speech typically consists of an alternation between vowel-like states (where the vocal tract is comparatively open and unobstructed) and consonant-like states where some obstruction to the airflow is made (thus altering speech between the two natural kinds of sounds). So, from the speech production point of view, a syllable consists of a movement from a constricted or silent state to a vowel-like state and then back to constricted or silent state. From the acoustic point of view, this means that the speech signal shows a series of peaks of energy corresponding to vowel-like states separated by troughs of lower energy (sonority).

Q. Explain Syllable

It can be divided into three possible parts as phonemes may occur at the beginning (onset), in the middle (nucleus or peak) and at the end (coda) of syllables - the combination of nucleus (peak) and coda is called the rhyme. The beginning (onset) and ending (coda) are optional while a syllable must have a nucleus (at least one phoneme). Thus, the study of the sequences of phonemes is called phonotactics, and it seems that the phonotactic possibilities of a language are determined by its syllabic structure (sequences of sounds that a native speaker produces can be broken down into syllables).

Q. Write a note on types of syllable.

A syllable structure could be of three types: ‘simple’ (CV), ‘moderate’ (CVC) and ‘complex’ (with consonant clusters at edges) such as CCVCC and CCCVCC (where V means vowel and C stands for consonant). Moreover, words can have one syllable (monosyllabic), two syllables (bisyllabic or disyllabic), three syllables (trisyllabic) or many syllables (polysyllabic).

Q. Explain stress as a suprasegmental feature

Stress is basically a suprasegmental feature applied to a whole syllable when it is made prominent by adding factors such as loudness, rise in pitch, length of duration and vowel quality (in

Phonetics and Phonology ENG507(Finals) 2nd Batch

contrast with other syllables). For example, in mi-ni-mi-**ZA**-tion, the second last (penultimate) syllable is prominent (as greater amount of energy is applied to it) as it is louder and longer than the rest of the syllables in the word. Similarly, its quality and pitch are different than that of others. All these factors make it stressed (compare: **IN**-sult vs. in**SULT** and be-**LOW** vs. **BILL**-ow).

Q. Write a note on types and categories of stress

One of the areas in which a little agreement is found is related to the levels of stress. Some descriptions of languages manage with just two levels (stressed and unstressed), while others use more than two. In English, one can argue that if one takes the word 'in-di-ca-tor' as an example, the first syllable is the most strongly (primarily) stressed one, the third syllable is the next most strongly (secondarily) stressed and the second and fourth syllables are weakly stressed or unstressed accordingly. This gives us three levels (primary, secondary and tertiary) and it is possible to argue for more, though this rarely seems to give any practical benefit.

Q. What are lexical and emphatic stress

In terms of its linguistic function, stress is often treated under two different headings: word (lexical) stress and sentence (emphatic) stress. Lexical stress is basically related to the primary stress applied at syllable level (when only one syllable is stressed) that has the ability to change the meaning and the grammatical category of a word as in the case of 'IMport' (noun) and 'imPORT' (verb). Sentence level stress, on the other hand, is applied on one word (rather than a syllable) in a sentence thus making that word more prominent (stressed) than the rest of the words in the sentence. This type of stress has its role in intonation patterns and rhythmic features of the language showing specific emphasis on the stressed word (which may be highlighting some information in the typical context).

Q. Explain stress timed languages

Languages of the world are; therefore, divided into two broad categories: stress timed language and syllable timed languages.

Stress timed languages have stress as their dominating rhythmic feature meaning that these languages seem to be timed according to the stressed patterns (the division among the syllables is made on the basis of stressed and unstressed patterns e.g., English and German languages). In other words, in stress timed languages, stressed syllables occur with regular intervals and their units of timing are perceived accordingly. Stress-timed rhythm is one of these rhythmical types, and is said to be characterized by a tendency for stressed syllables to occur at equal intervals of time. This idea is further clarified in the next topic.

Q. Give detail of syllable timed languages

In syllable timed languages, all syllables tend to have an equal time value (for example, their length or duration) and the rhythm of the language is said to be syllable-timed. In these languages, syllables tend to occur at regular intervals of time with fixed word stress. A classic example is Japanese in which all morae have approximately the same duration. This tendency is contrasted with stress-timing where the time between stressed syllables is said to tend to be equal irrespective of the number of unstressed syllables in between. Czech, Polish, Swahili and Romance languages (e.g., Spanish and French) are often claimed to be syllable-timed.

Q. Explain pitch as a suprasegmental feature

As a suprasegmental feature, pitch is an auditory sensation - when we hear a regularly vibrating sound such as a note played on a musical instrument (or a vowel produced by the human voice), we hear a high pitch (when the rate of vibration is high) and a low pitch (when the rate of vibration is low). There are some speech sounds that are voiceless (e.g. /s/), and cannot give rise to a sensation of pitch

Phonetics and Phonology ENG507(Finals) 2nd Batch

in this way but the voiced sounds can. Thus the pitch sensation that we receive from a voiced sound corresponds quite closely to the frequency of vibration of the vocal folds. However, we usually refer to the vibration frequency as fundamental frequency in order to keep the two things distinct. In tonal languages, pitch is used as an essential component of the pronunciation of a word and a change of pitch may cause a change in meaning. In most languages (whether or not they are tone languages) pitch plays a central role in intonation. In very simple words, pitch is the variation in the vibration of vocal folds.

Q. Explain in detail tone and tonal languages

Tone (in phonetics and phonology) as a suprasegmental feature refers to an identifiable movement (variation) or level of pitch that is used in a linguistically contrastive way. In tone (tonal) languages, the linguistic function of tone is to change the meaning of a word. For example, in Mandarin Chinese, [ma] said with a high pitch means 'mother' while [ma] said on a low rising tone means 'hemp'. In other (non-tonal) languages, tone forms the central part of intonation, and the difference between, for example, a rising and a falling tone on a particular word may cause a different interpretation of the sentence in which it occurs. In the case of tone languages, it is usual to identify tones as being a property of individual syllables, whereas an intonational tone may be spread over many syllables. In the analysis of English intonation, tone refers to one of the pitch possibilities for the tonic (or nuclear) syllable. For further analysis, a set of four types of tone is usually used (fall, rise, fall-rise and rise-fall) though others are also suggested by various experts.

Q. Write a note on intonation as a suprasegmental feature

In simple sense, 'intonation' refers to the variations in the pitch of a speaker's voice used to convey or alter meaning (at sentence level). In its broader and more popular sense, it is used to cover much the same field as 'prosody', where various features such as voice quality, tempo and loudness are also included. It is a term frequently used in the study of suprasegmental phonology, referring to the distinctive use of patterns of pitch, or melody and the study of intonation is sometimes called intonology. In some approaches, the pitch patterns are described as contours and analyzed in terms of levels of pitch as pitch phonemes and morphemes while in others, the patterns are described as tone units or tone groups which are further analyzed as contrasts of nuclear tone, tonicity, etc. The three variables of pitch range, height and direction are generally distinguished.

Q. What are functions of intonations

Intonation as a suprasegmental feature performs several functions in a language. Its most important function is to act as a signal of grammatical structure (e.g., creating patterns to distinguish among grammatical categories), where it performs a role similar to punctuation (in written language). It may furnish far more contrasts (for conveying meaning). Intonation also gives an idea about the syntactic boundaries (sentence, clause and phrase level boundaries). It also provides the contrast between some grammatical structures (such as questions and statements).

The role of intonation in the communication is quite important as it also conveys personal attitude (e.g., sarcasm, puzzlement, anger, etc.). Finally, it can signal contrasts in pitch along with other prosodic and paralinguistic features. It can also bring variation in meaning and can prove an important signal of the social background of the speakers.

Q. Discuss linguistic phonetics

Linguistic phonetics is an approach which is embodied in the principles of the International Phonetic Association (IPA) and in a hierarchical phonetic descriptive framework that provides certain basis for formal phonological theory. Speech, being very complex phenomena and having multiple levels of organization, needs to be explored from different angles. Linguistic phonetics answers a number of questions related to the possible ways of articulatory unified phonetics and phonology and from the

Phonetics and Phonology ENG507(Finals) 2nd Batch

perspective of cognitive phonetics focusing on speech production and perception and how they shape languages as a sound systems. The idea is mainly related to the overall ability of human beings to produce sounds (as a community and irrespective of their specific languages) and then the representation of their shared knowledge (as considered by the IPA in its charts) for formal phonetic and phonological theories.

Q. Explain phonetics of the community and of the individual

The description of the phonetics of the individual involves describing the phonetic knowledge and skills related to the performance of language. It is possible that certain aspects of the phonetics of the individual can be captured using IPA transcription but others are not compatible with it (such as his private knowledge and its performance and the role of memory and experience). Secondly, the phonetics of the individual is usually not the focus of the linguist in speech elicitation, and it is difficult to describe even with spectrograms of the person's speech. Although, the phonetics of the individual is the focus of much of the explanatory power of phonetic theory but for general phonetic description we need to focus on the phonetics of the community.

Q. Describe the IPA

IPA is the set of symbols and diacritics that have been officially approved by IPA. The association publishes a chart comprising of a number of separate charts. At the top inside the front cover, you will find the main consonant chart. Below it is a table showing the symbols for nonpulmonic consonants, and below that is the vowel chart. Inside the back cover is a list of diacritics and other symbols, and a set of symbols for suprasegmental features (events) such as tone, intonation, stress, and length.

Q. Explain in detail feature hierarchy

Feature hierarchy is an important concept in phonetics and phonology which is based on the properties and features of sounds. In a very general sense, a feature may be tied to a particular articulatory maneuver or acoustic property. For example, the feature [bilabial] indicates not only that the segment is produced with lips but also that it involves both of them. Such features (in phonetics and phonology) are listed in a hierarchy with nodes in the hierarchy defining ever more specific phonetic properties. For example, sounds are divided in terms of their supra-laryngeal and laryngeal characteristics, and their airstream mechanism. The supra-laryngeal characteristics can be further divided into those for place (of articulation), manner (of articulation), the possibility of nasality, and the possibility of being lateral. Thus, these features are used for classifying speech sounds and describing them formally.

Q. Name feature hierarchy

The four features (i.e., Stop, Fricative, Approximant, and Vowel) depend on the degree of closure of the articulators.

The manner category 'Stop' has only one possible value (i.e., [stop]), but 'Fricative' has two (i.e., [sibilant] and [nonsibilant]). Similarly, 'Approximant' and 'Vowel' have five principal features: Height, (with five possible values [high], [mid-high], [mid], [mid-low], and [low]), Backness (with three values, [front], [center], and [back]) and two kinds of 'Rounding' (i.e., Protrusion with possible values [protruded] and [retracted], and Compression, with possible values [compressed] and [separated]). The fourth feature for vowels and approximants is the 'Tongue Root' which has two possible values: [+ATR] and [-ATR]. Finally, the feature 'Rhotic' has only one possible value, [rhotacized]. It is also important to remember that the 'Laryngeal' possibilities involve mainly five features ([voiceless], [breathy voice], [modal voice], [creaky voice], and [closed] - forming a glottal stop). Airstream features have three values; Pulmonic, Velaric and Glottalic).

Q. Discuss controlling articulatory movements

Phonetics and Phonology ENG507(Finals) 2nd Batch

Focusing on the phonetics of the individual we can explore the controlling of the articulatory movement. For example, while underlying our linguistic description of [p], as an example of speech motor control, an array of muscular complexity involving dozens of muscles in the chest, abdomen, larynx, tongue, throat, and face is in action. Interestingly, all of these are contracted with varying degrees of tension in specific sequence and duration of contraction. For this sound (i.e., [p]), in order to produce a lip closure movement, two main muscles (depressor labii inferior and incisivus inferior) are activated to create 'too much' and 'enough tension'. Then at the same time the jaw muscles are activated so that it may trade with the lip muscles (together with lower lip movement) for closing and the opening. This structure specifies an overall task "close the lips" at the top node, and subtasks such as "raise the lower lip" and "lower the upper lip" are coordinated with each other to accomplish the overall task. Some subtasks also require further reduction of the goal into smaller subtasks. In addition to create a voiceless bilabial (i.e. [p]), the glottis needs to be wide apart (for free air passage). So exploiting the air passage, keeping it voiceless at larynx and creating a closure by lips and jaws along with many subtasks are achieved for [p] mainly because of the controlling articulatory movements thus enabling us to understand the individual variation in the production of speech.

Q. Explain Memory for Speech

The role of the memory for speech under the exemplar theory suggests that many instances of each word are stored in memory and their phonetic variability is memorized rather than computed. The main postulates of the concepts are given here:

- Language universal features
- Speaking styles
- Generalization and productivity
- Sound change

Q. What is the balance between phonetic forces

To produce sounds with maximum ease of articulation, only similar sounds are affected. The focus of the speakers is always on maintaining a sufficient perceptual distance between the sounds that occur in a contrasting set (e.g., vowels in stressed-monosyllabic words *beat*, *bit*, *bet*, and *bat*). This principle of perceptual separation does not usually result into one sound affecting an adjacent sound (as explained in the principle of maximum ease of articulation). Instead, perceptual separation affects the set of sounds that potentially can occur at a given position in a word, such as in the position that must be occupied by a vowel in a stressed monosyllable as in words *beat*, *bit*, *bet*, *bat* so that the perceptual separation is maximized. The principle of 'maximum perceptual separation' also accounts for some of the differences between languages. All these examples illustrate how languages maintain a balance between the requirements of the speaker and those of the listener.

Q. Define syllable

Syllable is the unit of pronunciation typically larger than a single sound and smaller than a word. A word maybe divided into parts such as in *ne-ver-the-less*.

Q. Define syllabification

Syllabification is the term which refers to the division of a word into syllables.

Q. Define resyllabification

Resyllabification refers to a reanalysis which alters the location of syllable boundaries.

Phonetics and Phonology ENG507(Finals) 2nd Batch

Q. Define monosyllabic

A word containing a single syllable is called a monosyllabic word.

Q. Define polysyllabic

If a word contains more than one, it is called polysyllabic.

Q. Explain Syllabification

There are different modes and structures for syllable structure and languages are labelled as per their syllabic templates. Consonant sequences are called clusters (e.g., CC – two consonants or CCC – three consonants). Most of the phonotactic analyses are based on the syllable structures and syllabic templates. On the basis of these consonant clusters, mainly three types of syllabic patterns are considered among languages; simple – moderate – complex (on the basis of consonants clusters at edges: onset and coda).

Q. Define phonotactics

The study of the phonemes and their order found in the syllables (the study of sound sequences) of a language is called the phonotactics.

Q. Explain Phonotactics

Phonotactics is a term used in phonology to refer to the order (sequential arrangements or tactic behavior) of segments (sounds or phonological units) which occur in a language. It shows us what counts as a phonologically well-formed structure of a word. The allowed sound patterns and restricted sound patterns of language are found through phonotactics. For example, in English, consonant sequences such as /fs/ and /spm/ do not occur initially in an English word, and there are many other restrictions on the possible consonant+vowel combinations which may occur. By thoroughly analyzing the data, the ‘sequential constraints’ of a language can be stated in terms of phonotactic rules.

Q. What is PRAAT? Discuss

Nowadays most of the research works in phonetics and phonology are based on software like Praat and WaveSurfer. Praat is a freeware created by Paul Boersma and David Weenink at the Institute of Phonetics Sciences of the University of Amsterdam. It is freely downloadable with improvised version and its guides and discussions are also available. One of the active platforms available for Praat related discussion and blogs is the [yahoopraatgroup](#). There is an introductory tutorial also available at the homepage of Praat. In short, Praat is a computer program with which you can analyze, synthesize, and manipulate speech, and create high-quality pictures for your articles and thesis.

Q. Discuss PRAAT Manual

The manual which we are going to use in this course was developed by the faculty (Sonya Bird and Qian Wang) at University of Victoria Canada. It is an excellent manual with worksheets and carefully designed ten labs. Praat only requires a very modest experience of computer and you can easily conduct experiments if you are using personal computer at a very initial level. It is very important for learning phonetics and phonology in general and acoustic analysis of speech in particular and will help you a lot in your future research work in the area of phonetics and phonology especially acoustic phonetics.

Q. Discuss recording and displaying on PRAAT

Phonetics and Phonology ENG507(Finals) 2nd Batch

- Go to NEW > RECORD mono-sound (with sampling rate – 44100 Hz)
- Make sure the volume bar is fluctuating as you record – if it isn't, you're not recording; if you don't see the volume bar at all, you're not speaking loudly enough.
- Watch out for clipping. If your recording level is too high and you go into the red on the volume bar, you'll end up with what is called a "clipped" signal; this is very bad for speech analysis!
- Give the recording a name (in the box below "Save to list").
- Save to list.
- Open the sound file in the Edit window.
- In the Objects window, highlight the sound file you've just recorded.
- Click Edit on the "Analysis and synthesis tools" panel.

As a final step of your Lab 1, you're now required to EXTRACT from this file, so:

- Select the portion that you want to study.
- File > Extract Selection (The extracted selection will show up as an entry ('sound untitled') in the Praat objects window).
- Close Edit window.

Q. Discuss segmenting and labeling

'Segmenting and labelling' is experiment on Praat which is particularly helpful in acoustic analysis because we will be able to place segmental symbols (label and add the textgrids to spectrogram) and annotate the sound file. The sound file needs text for a number of reasons including keeping the track record of your measurement from the file for annotating the textgrid tiers and by learning segmenting and labelling, we also learn how to open both files together and putting the phonetic symbols to our recordings.

Q. Describe various steps in learning segmenting and labeling on Praat

Follow the following steps for learning segmenting and labeling on Praat:

1. Create a textgrid:
 - In the Praat Objects window, highlight the subject (required) file.
 - Annotate > To TextGrid.
 - Create two tiers (this will be enough for our purposes). Write 'word segment' (these are two tiers) on the cell named 'All tier names' on the small window.
2. Open the sound file and textgrid together:
 - Hold down Ctrl and click on each file to highlight them both.
 - Edit (in your display you should now see the waveform (top), the spectrogram (middle) and the textgrid (bottom) corresponding to your sound file).
3. Segment the file:
 - Place the cursor at the beginning of the name on the spectrogram/waveform; a boundary line will show up.
 - Click in the little circle at the top of the word tier in the Textgrid to create a boundary.
 - To remove a boundary that you have made - Highlight the boundary - Go to **Boundary > Remove** OR click Alt+backspace.
4. Label the intervals:
 - Select/highlight the target interval by clicking between two boundaries; the selected interval should go yellow.

Phonetics and Phonology ENG507(Finals) 2nd Batch

- To input or change the text in an interval, edit in the Textbox above the spectrogram.
- Give name to each interval you create ([first name] or [last name]).

Q. How you will export visual display to word file

Exporting your visual displays to the Word document is the part of the ‘write up’ of the experiment. It is an important step as we need to report every experiment to our Word file for reporting purposes. There is quite an easy way to do it by maximizing the edit window and hitting the ‘PrtScr’ (print screen) button and, subsequently, pasting it into Word document.

1. Export your labeled waveform and spectrogram to Word:
 - Maximize the Edit window: click on the little square at the top right of your Edit window.
 - Hit PrtSc (“Print Screen”) on your keyboard - nothing would happen on your screen at this point.
 - Open Word document and provide an introduction to the image as required.
 - Hit Ctrl+V to paste the Praat image into your Word document.
 - Give your Figure a number and title below the image.

Q. Discuss the source-filter model of speech

Source-filter theory is particularly important to understand the basic components of speech sounds and the nature of the acoustic signals (it is the physics of speech sounds). It is, therefore, very crucial to understand the acoustic analysis of speech sounds particularly the vowels and vowel-like (sonorous sounds). The major goal of this lab is to understand and explore the basic acoustic components of (sonorous) speech sounds such as fundamental frequency, harmonics, and formants. Basically, these components create the acoustic signals associated with speech – understanding them is crucial to understanding what we actually hear, when we hear speech sounds. The labs based on the source-filter theory are conducted in the next few sessions.

Q. Discuss in detail measuring the fundamental frequency

Fundamental frequency (F0 or Pitch) is an important component of source-filter theory. It can be taken from the middle of the sound (e.g., vowel) using three techniques including displaying the Pitch track (automatically), measuring from Pitch track manually, and by looking at the waveform – by highlighting one cycle. Before measuring the fundamental frequency, we need to have a recorded file containing a vowel sound (e.g., [æ]). Let’s learn all of the three methods one by one:

1. Displaying the pitch track and allowing Praat to measure the pitch automatically:
 - Display the pitch track: Pitch > Show pitch.
 - Place your cursor in the middle – a stable portion of the vowel.
 - Go to Pitch > Get pitch – a box will appear with the pitch value in it (note it down)
2. Displaying the pitch track and measuring pitch manually:
 - Display the pitch track - Pitch > Show pitch.
 - Click on the blue pitch track in the middle of the vowel.
 - A red horizontal bar should appear with the pitch value (in dark blue) on the right side of the window (take the measurement from here).
3. By looking at the waveform (top of the display):
 - Zoom into a small piece of the waveform in the middle of the vowel and measure the period by highlighting one complete cycle and noting the time associated with it (in the panel above the waveform).

Phonetics and Phonology ENG507(Finals) 2nd Batch

Q. Discuss how to measure the Harmonics

Harmonics are the multiple integers of the fundamental frequency which are basically the result of vocal fold vibration (complex wave). We need the 'narrow band spectrogram' for measuring the H (which we can set by fixing the spectrum setting at 0.025). By starting measuring the frequency of the first three harmonics, we will go to the H10 (H1, H2, H3 – H10). Finally, we will compare with pitch measurement already taken. It is important to note down that when our vocal folds vibrate, the result is a complex wave, consisting of the fundamental frequency plus other higher frequencies, called harmonics. Let's now take the harmonics:

1. Display a narrow-band spectrogram:
 - Go to: Spectrum > Spectrogram settings.
 - Change the window length to 0.025s – the default window length is 0.005s (wide-band spectrogram) - this changes the spectrogram dramatically!
 - Looking at each vowel, notice the grey horizontal bands: these correspond to harmonics. For each vowel, measure the frequencies of the first 3 harmonics (H1-H3) and the 10th harmonic (H10).
 - Click on the center (horizontally) of each harmonic in the center of each vowel.
 - A red horizontal bar should appear with the frequency value on the left side of the window in red.

Q. Discuss how to measure the Formants

Formants are the overtone resonances. Acoustically, in order to plot vowels on chart, F1-F2 are very important. We need the wide bands for measuring the formants (which are the important characteristics of sonorant speech sounds – vowels). On spectrogram, formants are thick bands (darkness corresponds to loudness; i.e. the darkest harmonics are the ones that are the most amplified). These amplified harmonics form the formants that are characteristic of sonorant speech sounds. Now, let's measure the first and second formants (F1 and F2) from the middle of each vowel using the three techniques outlined below and note down your measurements:

1. Displaying the formants (red dots on the spectrogram) automatically:
 - Display the formant track: Formant > Show formants.
 - Place your cursor in the middle, stable portion of the vowel.
 - Go to Formant > Formant listing: a box will appear with the time point at which the measurement was taken, and the first four formants.
2. Displaying the formants and measuring the frequency manually:
 - Display the pitch track: Formant > Show formants.
 - Place your cursor in the center of each formant, in the middle of the vowel.
 - A red horizontal bar should appear with the frequency value on the left side of the window in red.
3. Measuring the frequency without displaying Praat formants – the easiest way – if Praat's formant tracking goes wonky:
 - Get rid of Praat's formant tracking: Formant > Show formants (unclick).
 - Place your cursor in the center of each formant, in the middle of the vowel.
 - A red horizontal bar should appear with the frequency value on the left (in red).

Q. What is relationship between harmonics and formants

Captured in the source-filter model of the speech, it is clear now (from the comparison of the two values – for formants and harmonics) that harmonic numbers are different for one type of sounds but

Phonetics and Phonology ENG507(Finals) 2nd Batch

the formants are the same. Actually, the relationship between the harmonics and the formants is captured in the source-filter model of speech production. The point is that harmonics are related to the laryngeal activity (source) and formants are the output of the vocal tract (filter).

Q. Discuss vowel properties

The eight vowels from American English are to be recorded for the purpose. These vowels are: heed, hid, head, had, hod, hawed, hood and who'd. Measure the following three things: intrinsic pitch, spectral make up (formants) and plot them in excel sheet (and finally export them to your Word document). Now, record yourself saying the words. Take a quick look at your vowels in the Edit window, and make sure you can clearly see the vowel formants. If you have trouble seeing them, you can go back to the previous labs and learn it again. While doing this, please make a note of it on your worksheet.

Q. Discuss in detail intrinsic pitch

Explore the eight vowels recorded in the last session and measure the pitch (F0) for each of them. You can measure pitch by using any of the three ways. Having measured the pitch (F0) in each of the vowels, note down your measurements. There is one more way to confirm your pitch measurement by looking at the spectral slice (which gives the component frequencies and their amplitudes).

- Select a portion of the vowel (70-80ms).
- Spectrum > View spectral slice.
- Click on the first (big) peak = H1 = F0 (Ignore any small spikes at the beginning; this might be noise). Now note down the frequency of this peak at the top of the vertical bar).

Use the confirmed pitch values and plot the pitch of each vowel on your excel sheet. Make sure you label your y-axis using a scale that allows you to spread out your measurements as much as you can. Now draw the cluster chart from the excel sheet and export to Word document and give the figure number and title.

Q. Discuss spectral make-up (Formant Values)

In order to have the spectral make-up of the vowels, we need to take the formant values (for first two formants i.e., F1 and F2) of the eight vowels already recorded. Calculate the first two formants for each vowel. This you can do by using the automatic formant tracking or the manual measurement. Having taken the values for all of the vowels, we will subsequently plot them on a chart. We need the default wide-band spectrogram for measuring the first and second formants (F1 and F2) of each vowel. You can also use Praat's automatic formant tracking to help you if you want. Also note and try to answer the following questions.

- a. Why do formants (F1 and F2) differ across vowels?
- b. What does F1 seem to correspond to, in terms of articulation?
- c. What about F2? To what does it correspond?

Q. How will you plot vowels on chart

By putting F1 and F2 in separate columns, write the formant values associated with different vowels (giving vowels in the first column, the difference between F2 and F1 in the second column and F1 in the third). After putting the data in Excel sheet, we will use the Scatter chart from the same spreadsheet. Further in order to make it corresponding with the required values for F1 and F2, we will reverse the values for both formants (on both axis – Y and X). Now the zero for both F1 and F2 is at the right corner. Once completed, export the chart to your Word document and give it the number and title accordingly.

Q. Explain sonorants and their formants

Phonetics and Phonology ENG507(Finals) 2nd Batch

Sonorants are vowel-like sounds (nasals and glides). These sounds are called sonorants because they have formants. But they are different from vowels because they generally have lower amplitude; therefore, they behave like consonants.

Q. Define nasal formants

Formants for nasal sounds are also important for acoustic analysis. Measure the first three (F1, F2 and F3) formants of nasals from the file. Nasals have very distinctive waveforms (different than that of vowels) as they have distinctive forms of anti-formants (bands of frequencies damped) and formant transition.

Q. Discuss glide and their formants

Glides are also the sonorants (vowel-like) sounds as they have similar patterns (have formants). Take the first three formants (F1, F2 and F3) from the middle of the sounds for glides (both for /w/ and /j/) and explore their acoustic correlates. Carefully judge the center of these sounds (the midpoint of [w] and [j]). Analyze that how similar is the formant structure of glides with vowels and nasals. Draw lines to indicate F1, F2, F3 and compare with vowels.

Q. Explain stop voicing on spectrograph

There are three important acoustic correlates of voicing in stops: the voice bar, VOT, and the duration of the preceding vowel. Record /apa/, /aba/, /ata/, /ada/, /ap^ha/ and /at^ha/ and for each of the stops in the file, take the three measurements. See the voicing or the voice bar by exploring features of stop. Also check the duration of the preceding vowels. Note down the presence of voicing.

Q. How to measure voice onset time (VOT)

It is the characteristic of voiced + voiceless + aspirates stop sounds and there are very easy steps to calculate the VOT. Record /apa/, /aba/, /ata/, /ada/, /ap^ha/ and /at^ha/. Zoom in through your stop sounds so that you can analyze the patterns of the stop sounds and find the difference among the three types of VOT (negative, zero and positive). Measure the VOT of each stop and compare voiced/voiceless counterparts (p/b, t/d, k/g). Similarly, zoom in so that you can clearly see the stop closure followed by the beginning of the vowel. You can measure the time between the end of the stop closure (the beginning of the release burst) and the onset of voicing in the following vowel (the onset of regular pitch pulses in the waveform). This is voice onset time or VOT.

Q. Write a comprehensive note on P&P research as the part of ELT

Phonetics and phonology is a very potential area for research to be carried out in Pakistani context. In applied phonology, many areas can be explored; for example, issues faced by Pakistani learners of English may be studied. Similarly, the pronunciation issues of Pakistani learners are potential area through which the difficulties faced by Pakistani students may be addressed. Also, researchers can explore and document the features of Pakistani English based on their phonological features in order to get the Pakistani variety of English recognized. Other problematic areas may also include: segmental and supra segmental features (such as stress placement, intonation patterns and syllabification and resyllabification of English words by Pakistani learners. Contrastive analysis (between English phonology and the sound systems of the regional languages of Pakistan (Urdu, Punjabi, Sindhi, Balochi and Pashto) can also be carried out by the researchers. We can also think about exploring the consonant clusters and interlanguage phonology from second language acquisition point of view. While focusing on ELT as the part of applied linguistics, studies may also be carried out on Pakistani variety of English (development of its corpora, deviation from the standard variety (RP), its specific features, etc.). Moreover, IPA resources and their application on ELT in Pakistani context can also be studied.

Phonetics and Phonology ENG507(Finals) 2nd Batch

Q. What are the current trends in P&P research

Regional languages of Pakistan may also be documented and studied. In this context, Pakistani researchers can get their work published (on the sounds – IPA illustrations) in international reputable journals such as the IPA Journal of Cambridge University. Pakistani regional languages are the part of the rich linguistic regions. (Himalaya Hindu Kush (HKH) region, one of the richest regions in the world linguistically and culturally) may be very potential area for research in the fields of areal and typological linguistics (description of linguistic features cross-linguistically). While working on Pakistani regional languages, one may apply for funding from international organizations (e.g., organization for endangered languages and UNISCO).

Q. Write a note on distinctive features

There are many other areas which may be explored. The distinctive features, for example, can be studied as the part of the phonetic studies of sounds (in applied phonology). Such a study would discover facts about the features of English pronunciation (and, of course, about the sounds of other languages). While working on this aspect of sound systems, the phonological analysis (theoretical) may include the description of phoneme as a combination of different features (e.g., /d/ as a phoneme and its features – alveolar, stop, voiced, oral and central) in a binary (+ -) order which is an important component of phonology. Moreover, the feature analysis may also include aspects of the target language as the part of ELT (English Language Teaching learning point of view). We need to include three principles for feature analysis: contrastive function (how it is different), descriptive function (what it is) and classificatory function (based on broader classes of sounds). Features may also be studied further as a part of language universals and then their role as language specific sub sets.

Q. Explain experimental phonetics

In experimental phonetics and phonology, the studies of sounds include various latest experimental techniques and computer software that are used under carefully designed lab experimentation. It is an important aspect of the application of the latest technology by going beyond the simple acoustics and by working in sophisticated phonetic labs in order to discover the hidden aspects of human speech. For example, questions such as ‘How speech is produced and processed?’ are the focus of experimental phonetics. The latest trends under experimental phonetics include brain functions in speech production and processing (by using the latest equipment – many special instruments such as x-ray techniques), speech errors, neurolinguistics and the topics related to the developments through computers – for speech analysis and synthesis.

Q. Discuss the study of variety

The varieties of English (or many other major languages such as Urdu, Punjabi and Pashto, for that purpose) may also be viewed as potential areas of research in the domain of phonetics and phonology. Such a study would include comparisons and contrasts (similarities and differences) among the accent (varieties) of the subject language(s) – e.g., differences at phonetic and phonological levels and the study of segmental and suprasegmental features – in different varieties. Differences among accents of English have already been discussed in this course under various headings (such as vowels in other accents of English). Similarly, English dialectology has already been explored by a number of studies with particular focus on the geographic differences (in the recognition of various forms of English – or Englishes). There are now many well-known data gathering techniques used in field linguistics (such as the sociolinguistics of English varieties) like variations paradigm. Field workers are particularly trained for data gathering and their expertise are developed for large scale studies related to the studies of varieties.

Q. Explain the relationship between ELT and P&P

Phonetics and Phonology ENG507(Finals) 2nd Batch

Phonetics and phonology is an integral part of English Language Teaching (ELT). The teaching of phonetics and phonology further needs to be integrated in the teaching of ELT. For the purpose, the teachers are expected to work on their skills related to the pronunciation of English and sensitize their students related to the topic – they may use their self-initiated procedures for carrying out the phonological contrastive analysis (CA) (e.g., of their mother tongues and English) at segmental and suprasegmental levels for enhancing their skills and completing their researches. They are also expected to take part in the phonology based ELT activities from the TESOL Home Page (available online) and participate in the English Language Teaching Reforms (ELTR) projects of the Higher Education Commission (HEC) of Pakistan and activities planned and sponsored by British Council Pakistan. The students are also expected to be the part of these platforms through their social groups and online learning opportunities.